## Summary of this week

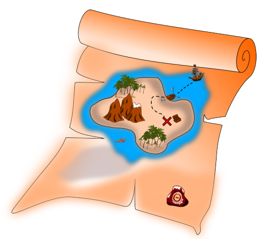
- This week, you learned:
  - How to refine the EKF via iteration for models having highly nonlinear measurement equations.
    - How to implement the IEKF in Octave code.
  - How to refine the SPKF when process and measurement noises are additive.
    - How to implement the CKF in Octave code.
  - How to estimate $\Sigma_{\widetilde{w}}$ and $\Sigma_{\tilde{v}}$ adaptively.
    - How to implement AEKF in Octave code.





## Where to from here?

- Next week, you will learn how to apply nonlinear Kalman filters to explore the key topic of adaptive model parameter estimation.
- You will learn how to use both SPKF and EKF to estimate the parameters of a model if the model's state vector is known.
  - This works very reliably.
- You will also learn two approaches to estimate states and parameters at the same time.
  - You will see that this can be tricky but it can also work well.







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